PROGRAMMING LABORATORY

Additional Test Cases and Reflection Questions

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CASE 1: NYERI–NAIROBI MATATU OPERATIONS

Test number	Input values	Expected behaviour	Actual result	Pass/Fail
1	Passengers = 12; Fare = KES 800; Parcels = 5; Fuel = 24 litres.	Program calculates revenue, expenses, occupancy and profit correctly.	Program calculates the expected values correctly.	Pass
2	Passengers = 0; Parcels = 5; Fare = KES 800.	Passenger revenue should be KES 0 and occupancy should be 0%. The program should identify that the trip did not make a profit.	Passenger revenue is KES 0 and occupancy is 0%. Profitability test returns False.	Pass
3	Passengers = 14; Vehicle capacity = 14.	Occupancy should be exactly 100%. The program should not report that passengers exceeded capacity.	Occupancy is 100% and passengers did not exceed capacity.	Pass
4	Passengers = 15; Vehicle capacity = 14.	Program should identify that the number of passengers exceeded vehicle capacity.	passengers_exceeded_capacity returns True.	Pass

CASE 2: M-PESA AGENCY BALANCE RECONCILIATION

Test number	Input values	Expected behaviour	Actual result	Pass/Fail
1	Opening cash = KES 90,000; Opening float = KES 110,000; Deposits = KES 75,500; Withdrawals = KES 62,300.	Program processes transactions in the correct order and calculates expected closing balances.	Transactions are processed correctly and balances are calculated.	Pass
2	Customer deposits = KES 0.	Cash should not increase due to deposits and electronic float should not decrease due to deposits.	Cash and electronic float remain unchanged during the deposit step.	Pass
3	Cash balance = KES 10,000; Customer withdrawals = KES 15,000.	Program should identify that there is not enough cash to process the withdrawal.	Cash balance becomes negative, indicating a shortage.	Pass
4	Actual closing balances equal expected closing balances.	Cash discrepancy and float discrepancy should both be exactly zero.	Both discrepancies are 0 and reconciliation tests return True.	Pass

CASE 3: DAIRY COOPERATIVE FARMER PAYMENT

Test number	Input values	Expected behaviour	Actual result	Pass/Fail
1	Morning milk = 18.5 litres; Evening milk = 14 litres; Rejection rate = 4%.	Program calculates total milk, rejected milk, accepted milk and net payment correctly.	Program calculates all values correctly.	Pass
2	Rejection rate = 0%.	No milk should be rejected. Accepted milk should equal total milk delivered.	Rejected milk is 0 litres and accepted milk equals total milk.	Pass
3	Rejection rate = 100%.	All milk should be rejected and accepted milk should be 0 litres.	Accepted milk becomes 0 litres.	Pass
4	Accepted milk = 30 litres; Quality score = 3.5; Loan balance = KES 10,000.	All three premium-supplier conditions are exactly satisfied, so the farmer should qualify.	Premium supplier test returns True.	Pass

CASE 4: UNIVERSITY ICT SERVICE WALLET

Test number	Input values	Expected behaviour	Actual result	Pass/Fail
1	Opening balance = KES 2,500; Top-up = KES 1,000; 75 black-and-white pages; 8 colour pages.	Program calculates fees, printing, binding, internet and remaining balance correctly.	Program calculates all values correctly.	Pass
2	Remaining balance = KES 500.	Emergency balance should be considered preserved because the balance equals the minimum required amount.	emergency_balance_preserved returns True.	Pass
3	Scholarship eligibility = 0.	No scholarship discount should be applied to printing costs.	Scholarship discount is KES 0.	Pass
4	Remaining balance is enough for exactly 100 black-and-white pages.	Program should return True for the ability to print 100 additional pages.	can_print_100_pages returns True.	Pass

REFLECTION QUESTIONS

1. Which case required the most complex expression? Explain.

Case 2, the M-Pesa Agency Balance Reconciliation, required the most complex expressions because it involved processing several transactions in the correct order and calculating expected balances, actual balances, discrepancies and working capital.

2. Give one example where missing parentheses produced an incorrect result.

In Case 4, the expression available_balance - printing_cost + binding_cost + internet_cost produced an incorrect result because the binding and internet costs were added back instead of being deducted. Parentheses were needed to group all expenses together before subtracting them from the available balance.

3. What problem can occur when two integers are used in a division expression?

When two integers are divided in some programming languages, integer division may remove the decimal part of the answer. This can cause inaccurate results, especially in financial calculations and percentages. In Python, the / operator gives a decimal result, while // performs floor division.

4. Explain the difference between assignment and equality comparison.

Assignment uses = to give a value to a variable, for example, profit = 3000. Equality comparison uses == to check whether two values are equal, for example, profit == 3000. The first stores a value, while the second produces a Boolean result of True or False.

5. Explain the difference between simple assignment and compound assignment.

Simple assignment gives a value to a variable using =, while compound assignment performs an operation and updates the existing value. For example, fuel_price = fuel_price * 1.08 is a simple assignment expression, while fuel_price *= 1.08 is a compound assignment.

6. Why should financial calculations normally use decimal or floating-point values?

Financial calculations often involve percentages, fees and decimal amounts. Using decimal or floating-point values allows the program to represent fractional values instead of restricting calculations to whole numbers.

7. Identify one situation where a Boolean expression helped in decision-making.

In Case 3, Boolean expressions were used to determine whether a farmer qualified as a premium supplier. The farmer had to have at least 30 litres of accepted milk, a quality score of at least 3.5 and a loan balance not exceeding KES 10,000.

8. State one limitation of solving these cases using expressions alone without selection statements.

Expressions alone can calculate values and produce True or False results, but they cannot automatically perform different actions based on those results. Selection statements such as if, elif and else are needed to make the program respond differently to different conditions.